

COVER PAGE

IGRVT-50 Payload EE

Schematic

Version: 1.0.0



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: CoverPage.SchDoc

Sheet 1 of 15

Checked by: BAO BUI

Document No: 1

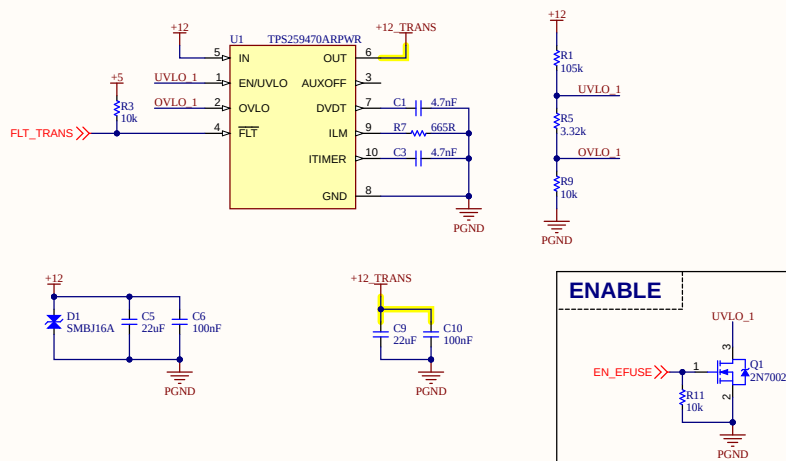
Size: A3

Approved by: VU PHAM

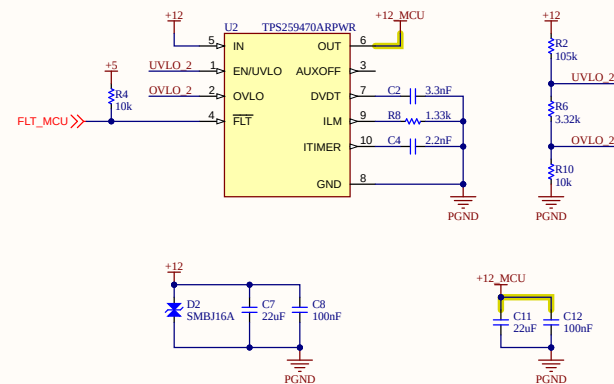
Revision: 1.0.0

Date: 8/1/2026

eFuse_Transducer



eFuse_MCU



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: eFuse.SchDoc

Sheet 2 of 15

Checked by: BAO BUI

Document No: 2

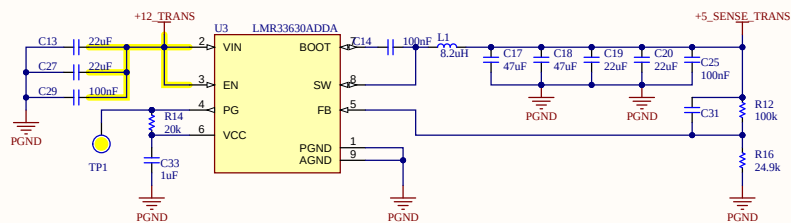
Size: A3

Approved by: VU PHAM

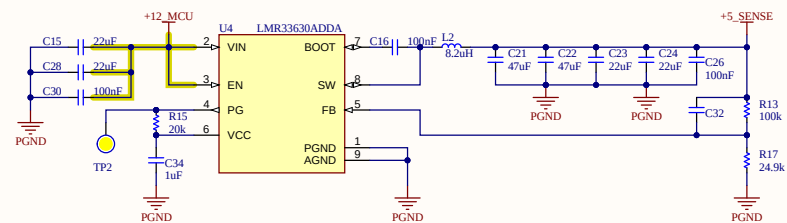
Revision: 1.0.0

Date: 8/8/2026

5V TRANSDUCER



5V MCU



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: Buck 5V.SchDoc

Sheet 3 of 15

Checked by: BAO BUI

Document No: 3

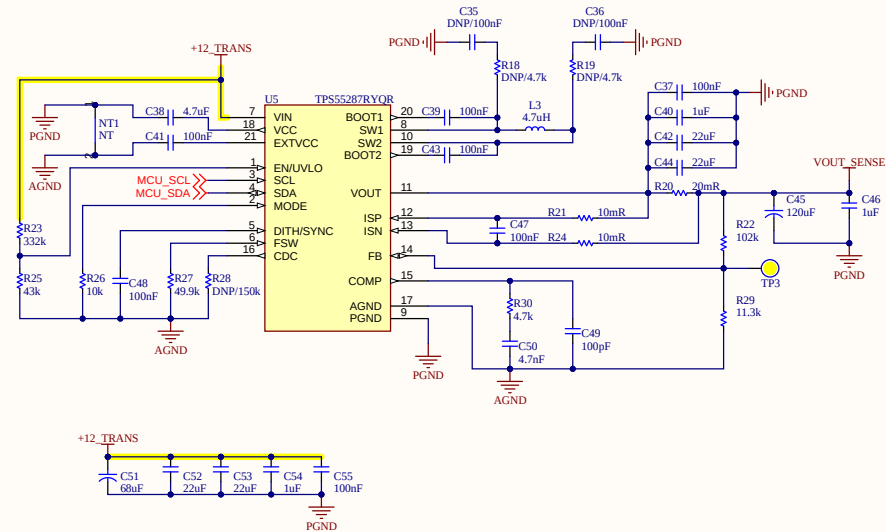
Size: A3

Approved by: VU PHAM

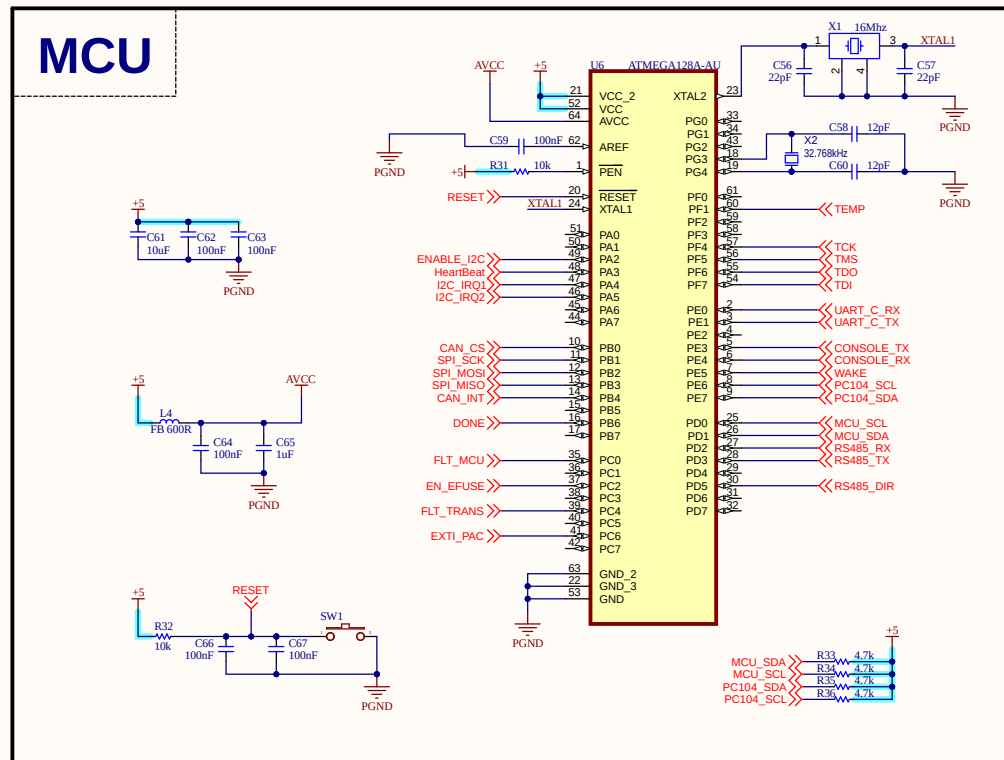
Revision: 1.0.0

Date: 8/4/2026

5-12V BUCK

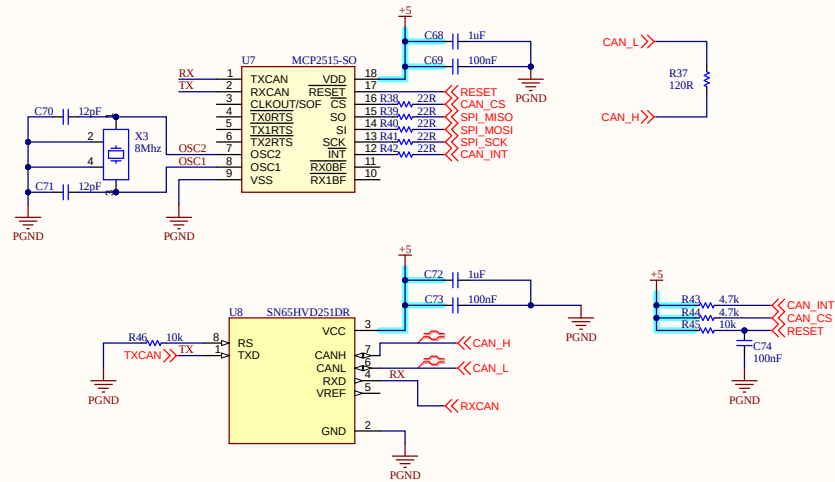


Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Buck 5-12V.SchDoc	Sheet 4 of 15
Checked by: BAO BUI	Document No: 4	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/4/2026



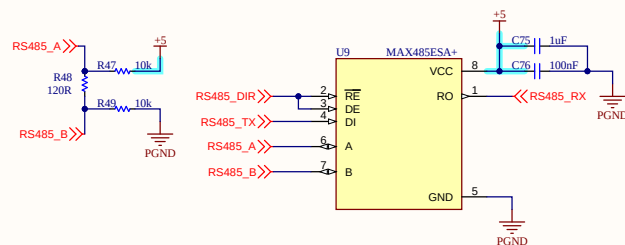
Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: MCU.SchDoc	Sheet 5 of 15
Checked by: BAO BUI	Document No: 5	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

CAN



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: CAN.SchDoc	Sheet 6 of 15
Checked by: BAO BUI	Document No: 6	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

RS485



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: RS485.SchDoc

Sheet 7 of 15

Checked by: BAO BUI

Document No: 7

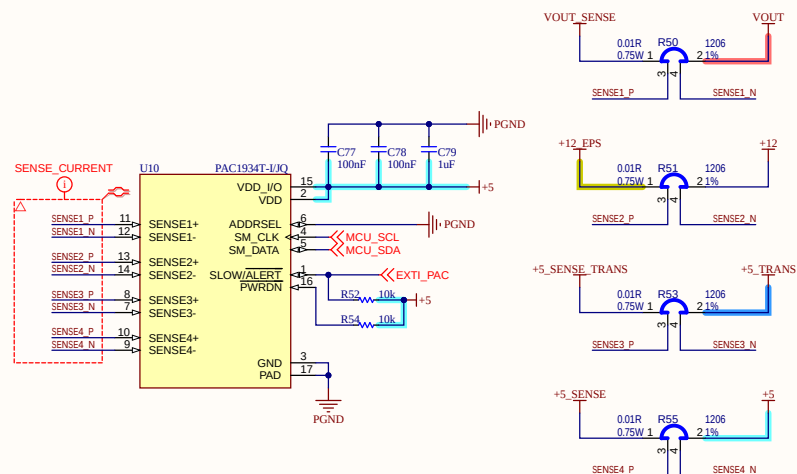
Size: A3

Approved by: VU PHAM

Revision: *

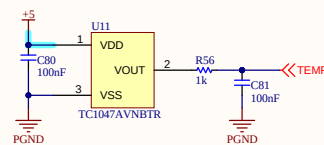
Date: 8/8/2026

POWER MONITORING



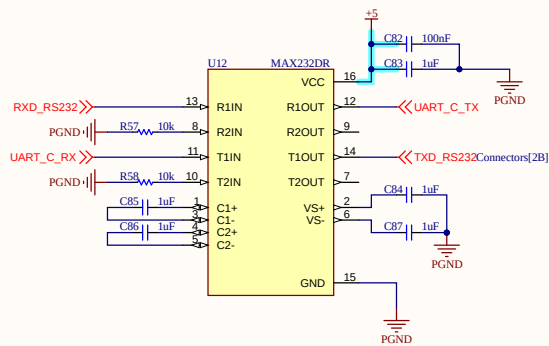
Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: PowerMonitor.SchDoc	Sheet 8 of 15
Checked by: BAO BUI	Document No: 8	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

TEMPERATURE SENSOR



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: TemperatureSensor.SchDoc	Sheet 9 of 15
Checked by: BAO BUI	Document No: 9	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

MAX232



Project: DAL_PDU

Sub-system: IGRVT-50

Design by: KIEN HOANG

Sheet title: MAX232.SchDoc

Sheet 10 of 15

Checked by: BAO BUI

Document No: 10

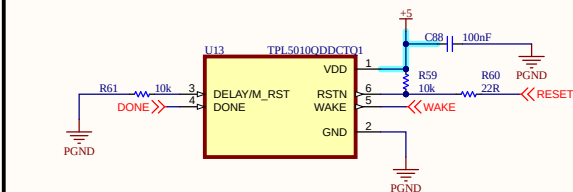
Size: A3

Approved by: VU PHAM

Revision: *

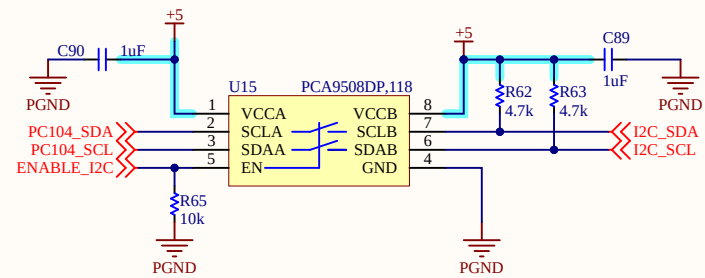
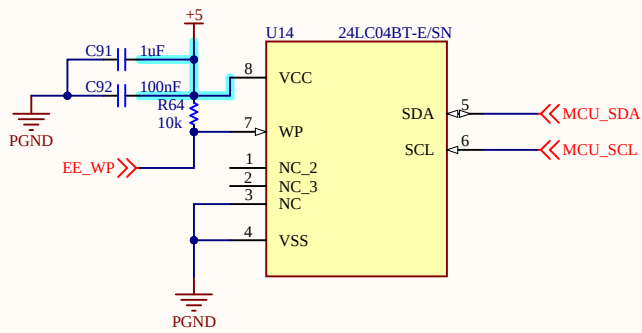
Date: 8/8/2026

WATCHDOG

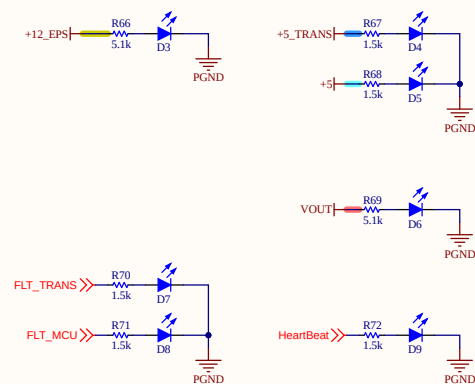


Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: WatchDog.SchDoc	Sheet 11 of 15
Checked by: BAO BUI	Document No: 11	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

I2C

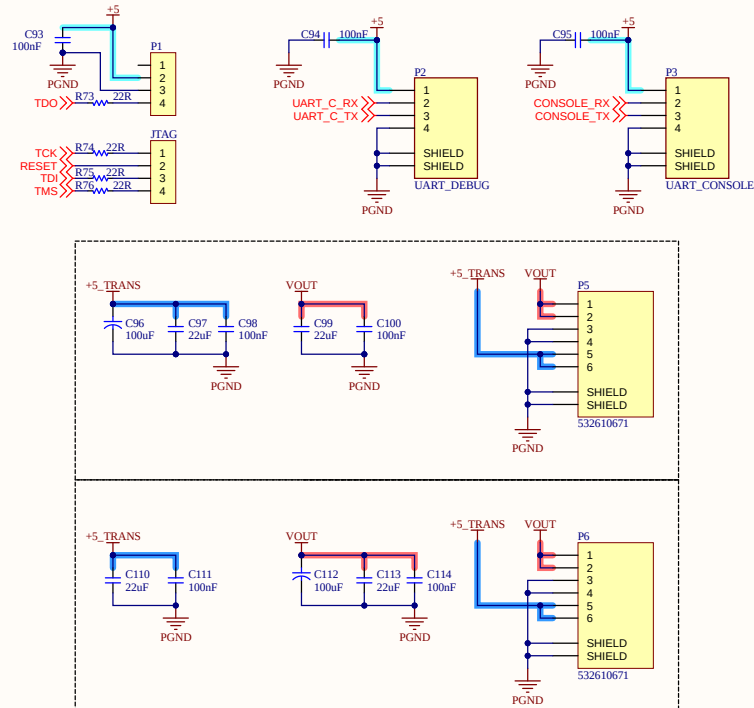
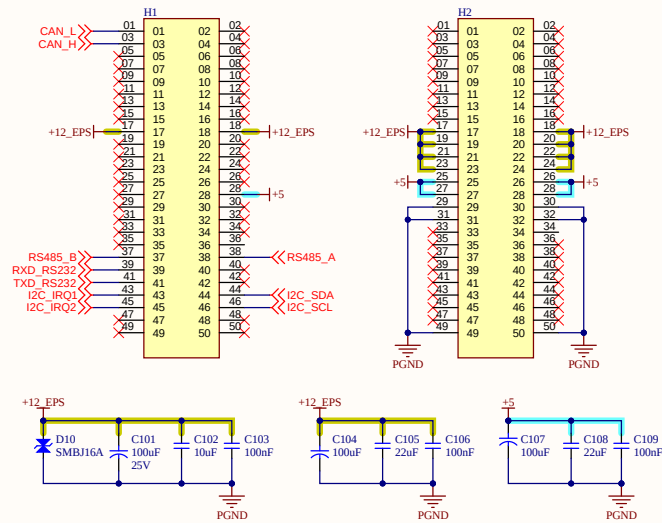


LED



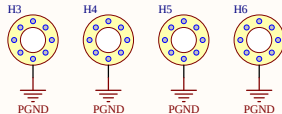
Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: LED.SchDoc	Sheet 13 of 15
Checked by: BAO BUI	Document No: 13	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

CONNECTOR



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Connectors.SchDoc	Sheet 14 of 15
Checked by: BAO BUI	Document No: 14	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/8/2026

HOLES



Project: DAL_PDU	Sub-system: IGRVT-50	
Design by: KIEN HOANG	Sheet title: Mounting Holes.SchDoc	Sheet 15 of 15
Checked by: BAO BUI	Document No: 15	Size: A3
Approved by: VU PHAM	Revision: *	Date: 8/1/2026

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	1k	RES	R56	1							
	1uF	CAP	C33, C34, C40, C46, C54, C65, C68, C72, C75, C79, C83, C84, C85, C86, C87, C89, C90, C91	18							
	1.32k	RES	R8	1							
	1.5k	RES	R67, R68, R70, R71, R72	5							
	2W7002	Mosfet N	Q1	1							
	2.2nF	CAP	C4	1							
	3.3nF	CAP	C2	1							
	3.32k	RES	R5, R6	2							
	4.7k	RES	R30, R33, R34, R35, R36, R43, R44, R62, R63	9							
	4.7nF	CAP	C1, C3, C50	3							
	4.7uF	CAP	C38	1							
	4.7uH	Inductor	L3	1							
	5.1k	RES	R66, R69	2							
	8MHz		X3	1							
	8.2uH	Inductor	L1, L2	2							
	10k	RES	R3, R4, R9, R10, R11, R26, R31, R32, R45, R46, R47, R49, R52, R54, R57, R58, R59, R61, R64, R65	20							
	10mR	RES	R21, R24	2							
	10uF	CAP	C81, C102	2							
	11.3k	RES	R29	1							
	12uF	CAP	C58, C60, C70, C71	4							
	16MHz		X1	1							
	20k	RES	R14, R15	2							
	20mR	RES	R20	1							
	22uF	CAP	C56, C57	2							
	22R	RES	R38, R39, R40, R41, R42, R60, R73, R74, R75, R76	10							
	22uF	CAP	C5, C7, C9, C11, C13, C15, C19, C20, C23, C24, C27, C28, C42, C44, C52, C53, C97, C99, C105, C108, C110, C113	22							
	24LC04BT-ESN		U14	1							
	24.9k	RES	R16, R17	2							
	43k	RES	R25	1							
	47uF	CAP	C17, C18, C21, C22	4							
	49.9k	RES	R27	1							
	68uF		T543X336M050A	1							
	100k	RES	C51	1							
			HE0407805	2							
	100nF	CAP	C6, C8, C10, C12, C14, C16, C25, C26, C29, C30, C31, C32, C37, C39, C41, C43, C47, C48, C55, C59, C62, C63, C64, C66, C67, C69, C73, C74, C76, C77, C78, C80, C81, C82, C88, C92, C93, C94, C95, C96, C100, C103, C106, C109, C111, C114	46							
	100uF	CAP	C49	1							
	100uF	Capacitor pol	C98, C101, C104, C107, C112	5							
	102k	RES	R22	1							
	105k	RES	R1, R2	2							
	120R	RES	R37, R48	2							
	120uF		T543X107M025A	1							
			HE090	1							
	332k	RES	R23	1							
	665R	RES	R7	1							
	S32G10671		PS, P6	2							
	ATMEGA1284-AU		U6	1							
	DNP14.7k	RES	R18, R19	2							
	DNP100nF	CAP	C35, C36	2							
	DNP150k	RES	R28	1							
	ESQ-126-14-G-D		H1, H2	2							
	FB 600R	BLM18KG601SN1D	L4	1							
	Header 4	Header, 4-Pin	JTAG, P1	2							
	Led	Typical LED	D3, D4, D5, D6, D7, D8, D9	7							
	LMR33630ADDA		U3, U4	2							
	MAX232DR		U12	1							
	MAX485ESA+		U9	1							
	MCP2515-SO	Stand-Alone CAN Controller with SPI Interface	U7	1							
	NT	Net Tie SMD	NT1	1							
	PAC1934T-IUQ		U10	1							
	PCA9508DP.118		U15	1							
	Shield		H3, H4, H5, H6	4							
	SMBJ11A	TVS Diode	D1, D2, D10	3							
	SN65HVD251DR		U8	1							
	SW2	Switch button 2 pin	SW1	1							
	TC1047AMNTR		U11	1							
	TP15010QDDCTQ1		U13	1							
	TPS55287RYQB		U5	1							
	TPS29470ARPWR		U1, U2	2							
	UART_CONSOLE		P3	1							
	UART_DEBUG		P2	1							
	Shunt 4 Pin	Shunt	R50, R51, R53, R55	4	Susumu	KRL1632E-M-R010-F-TS	Volume Production	DigiKey	408-1570-1-ND	0.43	1.72
	Crystal 2 Pins	Crystal	X2	1	NDK	NX20125A-32.768K-STD-MUB-1		Digi-Key	644-1219-1-ND		